

Machine learning (ML) is a subset of artificial intelligence that enables computer systems to learn from data and improve their performance over time without being explicitly programmed for specific tasks. Instead of following static, hard-coded rules, ML algorithms identify patterns within datasets to make predictions, classify information, or generate content.

The field is primarily categorized into three main types based on the learning approach:

Supervised Learning: Models are trained on labeled datasets (with known inputs and outputs) to predict outcomes for new data, commonly used for classification and regression.

Unsupervised Learning: Algorithms analyze unlabeled data to discover hidden structures, patterns, or groupings, such as clustering similar data points.

Reinforcement Learning: An agent learns to make decisions by interacting with an environment, receiving rewards or penalties to maximize long-term outcomes.

Modern ML serves as the backbone for many contemporary technologies, including deep learning (which uses multi-layered neural networks for complex tasks), generative AI (which creates new content like text and images), and various commercial applications such as recommendation engines, fraud detection, and natural language processing.